(Please write your Enrollment Number)

End-Term Examination
(CBCS) (SUBJECTIVE TYPE)
B. Tech CSAI, 4th Semester
(May, 2022)

Q.P Code 220560

Subject Code: BAI-204	Subject: Optimization Techniques and Decision Making

Time: 3 Hours **Maximum Marks:** 60

Note: Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1.	Classify each one of the following (parts a to f) as either True or False. All answers must be supported with a justification or counterexample as appropriate. Unsupported answers will be awarded zero marks.	**(2.5*8=20)**
	(a) If two different basic feasible solutions of a linear program are optimal, then they must correspond to adjacent vertices of the feasible region.	
	(b) When minimizing a convex function over a convex set, the optimal solution is always on the boundary of the set.	
	(c) The feasible set of an LP (Linear Programming) in standard form is always bounded.	
	(d) The feasible set of a semidefinite programming problem is always convex.	
	(e) A feasible linear program of minimizing c ' x subject to Ax ≥ b has a finite cost if and only if c is a nonnegative combination of the rows of A.	
	(f) The problem $\sum_{j=1}^{n} c_j\|x_j\|$ subject to $\sum_{j=1}^{n} a_j\|x_j\| \le b,$, with $c_j, a_j \ge 0,$ can be modeled as a linear optimization problem	
	(g) Define the role of Genetic Optimization in solving real-world problems.	
	(h) Define five real-world applications of the ant colony optimization algorithm.	
	UNIT-1	
Q2.	Define the role of optimization in many real-world mathematical problems. How constraints and feasible regions are the most important factors to optimize the problem in a better and more effective way?	**(10)**
Q3.	Define the way to provide a formulation of design problems as mathematical programming problems for optimization. Illustrate the different real-world applications of optimization in detail.	**(10)**
	UNIT-2	
Q4.	Consider the following transshipment problem: The supply nodes are A, B, the demand nodes are D, E, and the transshipment node is node C. There are four unknowns a, b, c, and d. The supply/demand amounts in the different nodes are: A: a, B: 400, D: −b, E: −200, whereas usual a positive amount indicates supply and a negative amount indicates demand. We are interested in finding an optimal (minimum cost) transshipment plan: (a) State conditions on a, b, c, d such that the above problem is feasible. (b) Consider the spanning-tree given by the edges {(A, D),(B, C),(C, D),(C, E)}. conditions on a, b, c, d for which the spanning tree solution will be feasible.	**(10)**
Q5.	You are planning on having access to a car for the next N years, where N is a fixed number. The price of a new car is P dollars. For reliability reasons, you will only own relatively new cars, at most m years old. The yearly cost of repairing and maintaining a car during its kth year is r_k, and it satisfies r1 < r2 < · · · < rm (i.e., it increases over time). At the end of any given year, you have the option of	**(10)**

	exchanging your k-year old car for a new one, with the corresponding trade-in value f_k of your old car satisfying t1 > t2 > · · · > tm (i.e., depreciating over time). You want to find the most economical sequence of buys and trade-ins, i.e., want to minimize the total cost over the N-year period. This includes all the money spent, either on buying or repairing (notice that you can sell your car at the end of the N years). (a) Propose a shortest path (or network flow) formulation for this problem.	
	UNIT-3	
Q6.	What type of algorithm is ant colony optimization (ACO)? What do you know about ant colony optimization present the steps? Explain the advantages of ant colony optimization. Illustrate your answer in detail.	**(10)**
Q7.	Illustrate the particle swarm optimization technique (PSO). Which is better PSO or ACO? How is velocity updated in particle swarm optimization? How is particle swarm optimization different from genetic algorithms? Illustrate your answer in detail.	**(10)**
	UNIT-4	
Q8.	What is meant by decision-making? Define the role of a decision-making tree in the case of classification or prediction? Illustrate the role of different models of decision making solving real-world problems.	**(10)**
Q9.	Define the concept of "Improving Decision Making Through Effective Knowledge Management" in your words. Explain general decision-making styles with the help of an example. Explain the advantages and disadvantages of group-aided decision-making. Give your answer in detail.	**(10)**